

11.24

(a)

```
##           Estimate Std. Error   t value   Pr(>|t|)
## (Intercept) -0.35956847 0.07915415 -4.5426355 1.501702e-05
## mon         -0.10824827 0.10338898 -1.0470000 2.975268e-01
## tue         -0.06609207 0.10103460 -0.6541528 5.144573e-01
## wed         -0.04927356 0.10377370 -0.4748174 6.359127e-01
## thu          0.03934669 0.10070504  0.3907122 6.968092e-01
## stormy       0.44633866 0.07921108  5.6348008 1.508139e-07
## mixed       0.23687528 0.07851343  3.0170036 3.209729e-03
```

since p-value of mixed = $3.209 \times 10^{-3} < 0.05 \Rightarrow$ we reject $H_0 (\pi_{\text{mixed}} = 0)$
and Hence it is statistically significant at the 5% Level

```
##
## Linear hypothesis test:
## stormy = 0
## mixed = 0
##
## Model 1: restricted model
## Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed
##
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     106 15.876
## 2     104 12.115  2    3.7604 16.14 7.857e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

F-value = 16.14 > 10
 $\rightarrow$ reject $H_0: \pi_{\text{stormy}} = \pi_{\text{mixed}} = 0$

(b)

```
##          Estimate Std. Error    t value    Pr(>|t|)
## (Intercept)  8.54023437  0.1566081  54.53250335  7.415812e-79
## lprice      -0.93014092  0.3533995  -2.63198161  9.768165e-03
## mon         -0.01190203  0.2083710  -0.05711941  9.545586e-01
## tue         -0.52583152  0.2022954  -2.59932461  1.068445e-02
## wed         -0.56261965  0.2069581  -2.71851907  7.673542e-03
## thu         0.09987059   0.2028454   0.49234827  6.235011e-01
## attr(,"df")
## [1] 105
## attr(,"nobs")
## [1] 111
```

TABLE 11.5 2SLS Estimates for Fish Demand					ln(price) :- 1.1194 → -0.9301, 價格彈性↑
Variable	Coefficient	Std. Error	t-Statistic	Prob.	
C	8.5059	0.1662	51.1890	0.0000	TUE, WED → is significant MON, THU → isn't significant
ln(PRICE)	-1.1194	0.4286	-2.6115	0.0103	
MON	-0.0254	0.2148	-0.1183	0.9061	
TUE	-0.5308	0.2080	-2.5518	0.0122	
WED	-0.5664	0.2128	-2.6620	0.0090	
THU	0.1093	0.2088	0.5233	0.6018	

(c)

##	Sargan	df	p.value	p-value = 0.3795 > 0.05 ⇒ do not reject H ₀ (IV is effective)
##	0.7722	1.0000	0.3795	

(d)

```
##
## Linear hypothesis test:
## mon = 0
## tue = 0
## wed = 0
## thu = 0
##
## Model 1: restricted model
## Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed
##
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
## 1	108	12.408				
## 2	104	12.115	4	0.29256	0.6279	0.6437

11.28

(a)

$$D: Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 PS_i + \alpha_4 DI_i + e_{di}$$

$$\Rightarrow \alpha_2 P_i = Q_i - \alpha_1 - \alpha_3 PS_i - \alpha_4 DI_i - e_{di}$$

$$\Rightarrow P_i = \frac{1}{\alpha_2} Q_i - \frac{\alpha_1}{\alpha_2} - \frac{\alpha_3}{\alpha_2} PS_i - \frac{\alpha_4}{\alpha_2} DI_i - \frac{1}{\alpha_2} e_{di}$$

'! $\alpha_2 < 0, \alpha_3 > 0, \alpha_4 > 0$, $\rightarrow$ the coef of $\begin{cases} Q_i < 0 \\ PS_i > 0 \\ DI_i > 0 \end{cases}$

$$S: Q_i = \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_{si}$$

$$\Rightarrow P_i = \frac{1}{\beta_2} Q_i - \frac{\beta_1}{\beta_2} - \frac{\beta_3}{\beta_2} PF_i - \frac{1}{\beta_2} e_{si} \quad \text{'! } \beta_2 > 0, \beta_3 < 0 \Rightarrow \text{coef of } \begin{cases} Q_i > 0 \\ PF_i > 0 \end{cases}$$

(b)

demand				
##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-11.428407	13.591606	-0.840843	4.081026e-01
## q	-2.670519	1.174955	-2.272869	3.153505e-02
## ps	3.461081	1.115572	3.102517	4.582228e-03
## di	13.389921	2.746707	4.874899	4.675236e-05
## attr(,"df")				
## [1]	26			
## attr(,"nobs")				
## [1]	30			

$Q < 0, PS > 0, DI > 0$
and all p-value < 0.05
 $\rightarrow$ significantly different from 0
(reject H_0)

supply				
##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-58.798223	5.8591611	-10.03526	1.316492e-10
## q	2.936711	0.2157717	13.61027	1.320944e-13
## pf	2.958486	0.1559638	18.96905	3.879687e-17
## attr(,"df")				
## [1]	27			
## attr(,"nobs")				
## [1]	30			

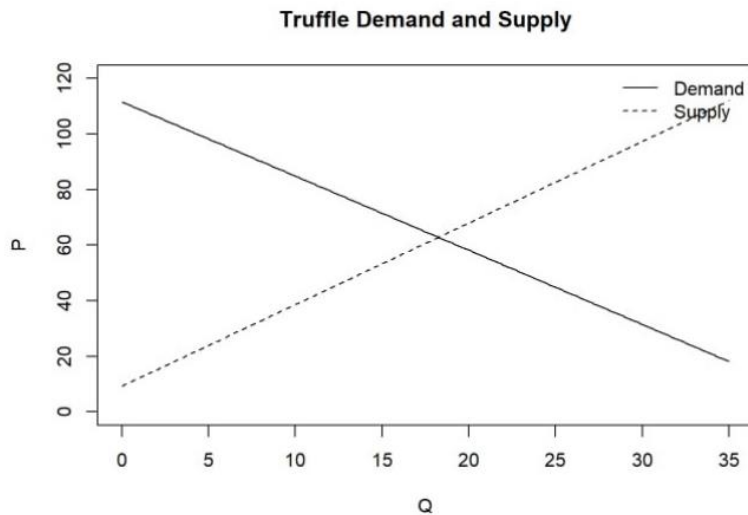
$q > 0, pf > 0$
and all p-value < 0.05
 $\rightarrow$ significantly different from 0
(reject H_0)

(c)

##	Pbar	Qbar	elasticity.q
##	62.7240	18.4583	-1.2725

(d)

##	D_intercept	D_slope_Q	S_intercept	S_slope_Q
##	111.5801	-2.6705	9.2470	2.9367



(e)

equilibrium values of P, Q in (d):

```
##      P_eq      Q_eq
## 62.8426 18.2502
```

predict equilibrium value:

```
##      Q_rf      P_rf
## 18.2604 62.8154
```

```
##              Q      P
## structural_2SLS 18.2502 62.8426
## reduced_form    18.2604 62.8154
## difference      -0.0102  0.0272
```

→ 兩者結果接近

(f)

supply

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -52.8763    5.0238  -10.5252      0
## q            2.6613    0.1712   15.5436      0
## pf           2.9217    0.1482   19.7150      0
```

$p\text{-value} < 0.05$

→ coef are significantly different from 0

Demand

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -13.6195    9.0872  -1.4987    0.1460
## q            0.1512    0.4988    0.3032    0.7642
## ps           1.3607    0.5940    2.2909    0.0303
## di           12.3582    1.8254    6.7701    0.0000
```

→ not significantly

] → significantly

compare to (b)

demand (b)

```
## q      -2.670519
## ps     3.461081
## di     13.389921
```

```
q      0.1512
ps     1.3607
di     12.3582
```

OLS x
OLS 估計
OLS 較小

supply (b)

```
q      2.936711
pf     2.958486
```

```
q      2.6613
pf     2.9217
```

OLS 較小
接近